



CS31007

**COMPUTER
ORGANIZATION
AND**

ARCHITECTURE

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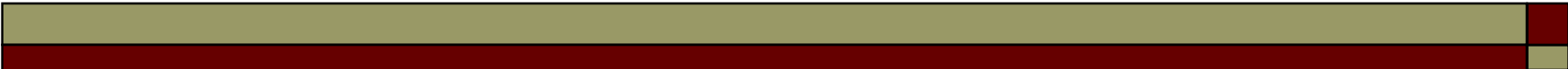
References/Text Books

□ **Theory:**

- ***Computer Organization and Design, 4th Ed, D. A. Patterson and J. L. Hennessy***
- ***Computer Organization, Hamacher, Vranesik and Zaky***
- ***Computer Architecture, Berhooz Parhami***
- ***Computer Architecture and Organization, J. P. Hayes***
- *Computer Architecture, Berhooz Parhami*
- *Microprocessor Architecture, Jean Loup Baer*

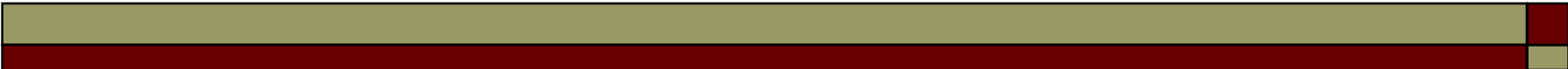
□ **Laboratory:**

- *Douglas Smith, HDL Chip Design (for Verilog)*
- *SPIM Tutorial :pages.cs.wisc.edu/~larus/spim.html*



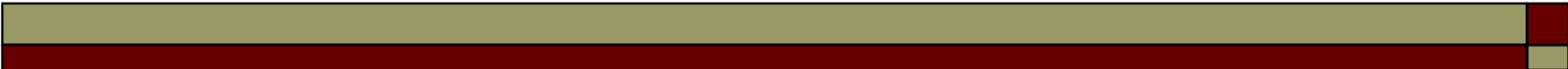
Detailed Syllabus

- See: <https://sites.google.com/view/coa2026/home>



Pre-Midsem Syllabus

- Overview: Terms and Taxonomy
- Instruction Set Architecture:
 - Instruction and Addressing
 - Procedures and Data
 - Assembly Languages Programs:
 - SPIM simulator and Debugger
- Numbers and Computers:
 - Number Representations
 - Adders
 - Multipliers
 - Floating Point Arithmetic

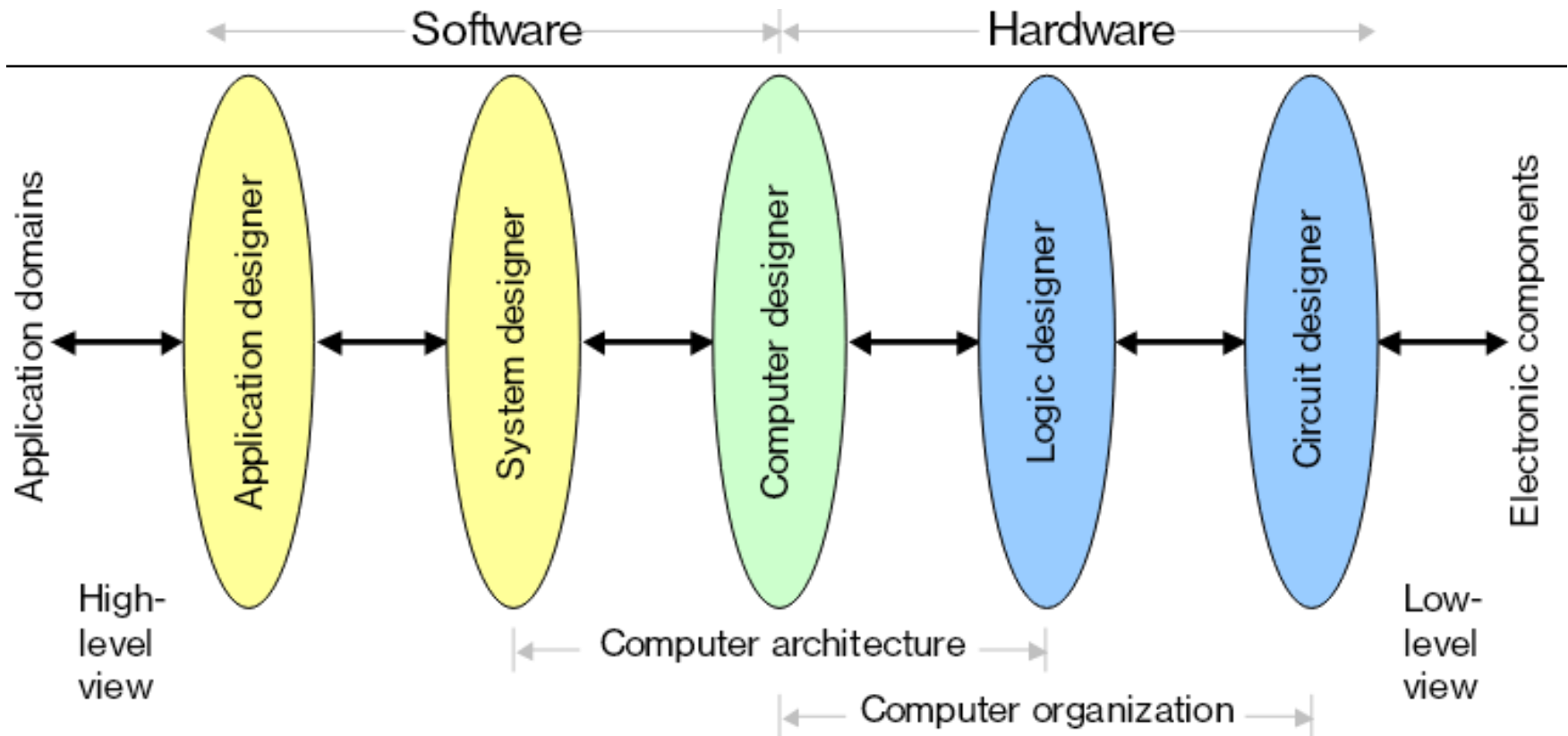


Pre-MidSem Syllabus

- Design of a Processor:
 - Instruction Execution Steps
 - Control Unit Design: Microprogramming
 - Pipelined Data Paths
 - Performance
 - Hazards
- This part will involve verilog (Hardware Description Language) coding

An Overview

3.1 From Components to Applications



Subfields or views in computer system engineering.

Difference between CO and CA

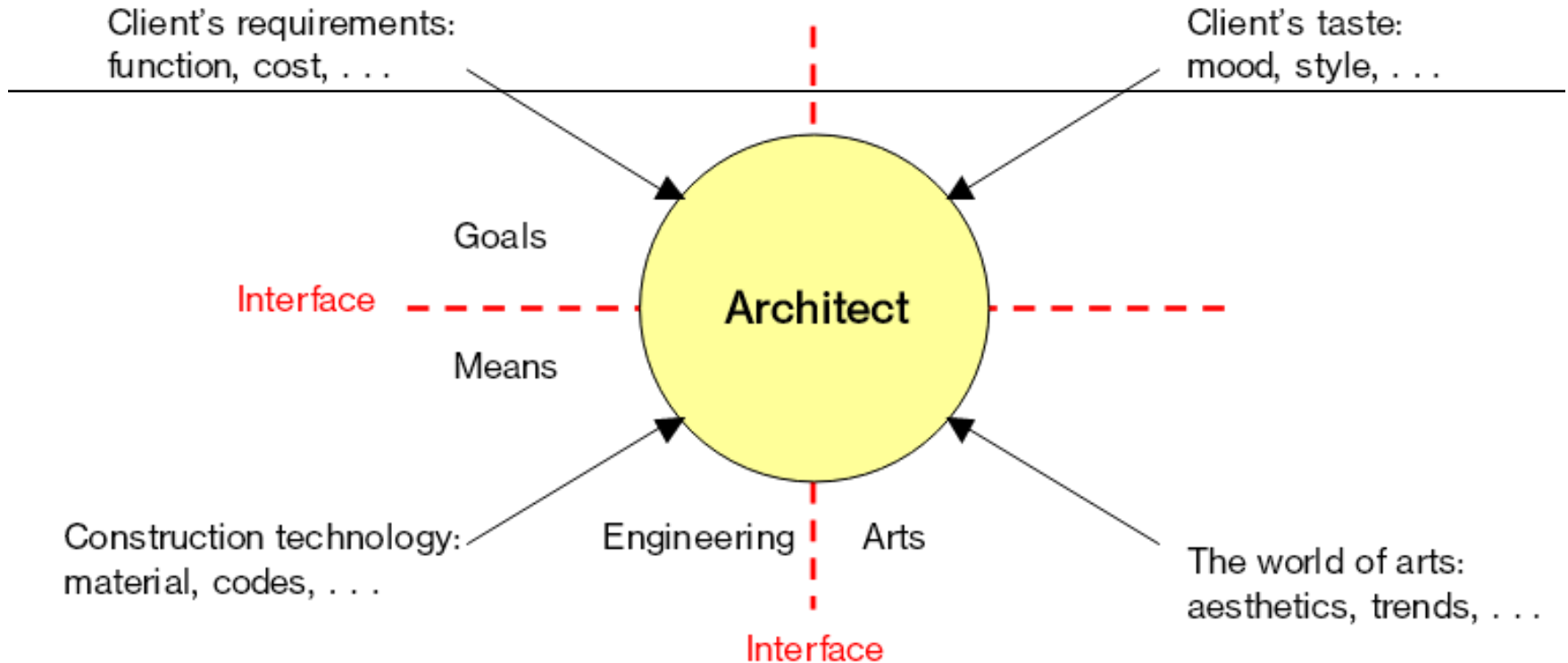
□ **Computer Organization:**

- Design of the components and functional blocks using which computer systems are built.
- **Analogy:** civil engineer's task during building construction (cement, bricks, iron rods, and other building materials).

□ **Computer Architecture:**

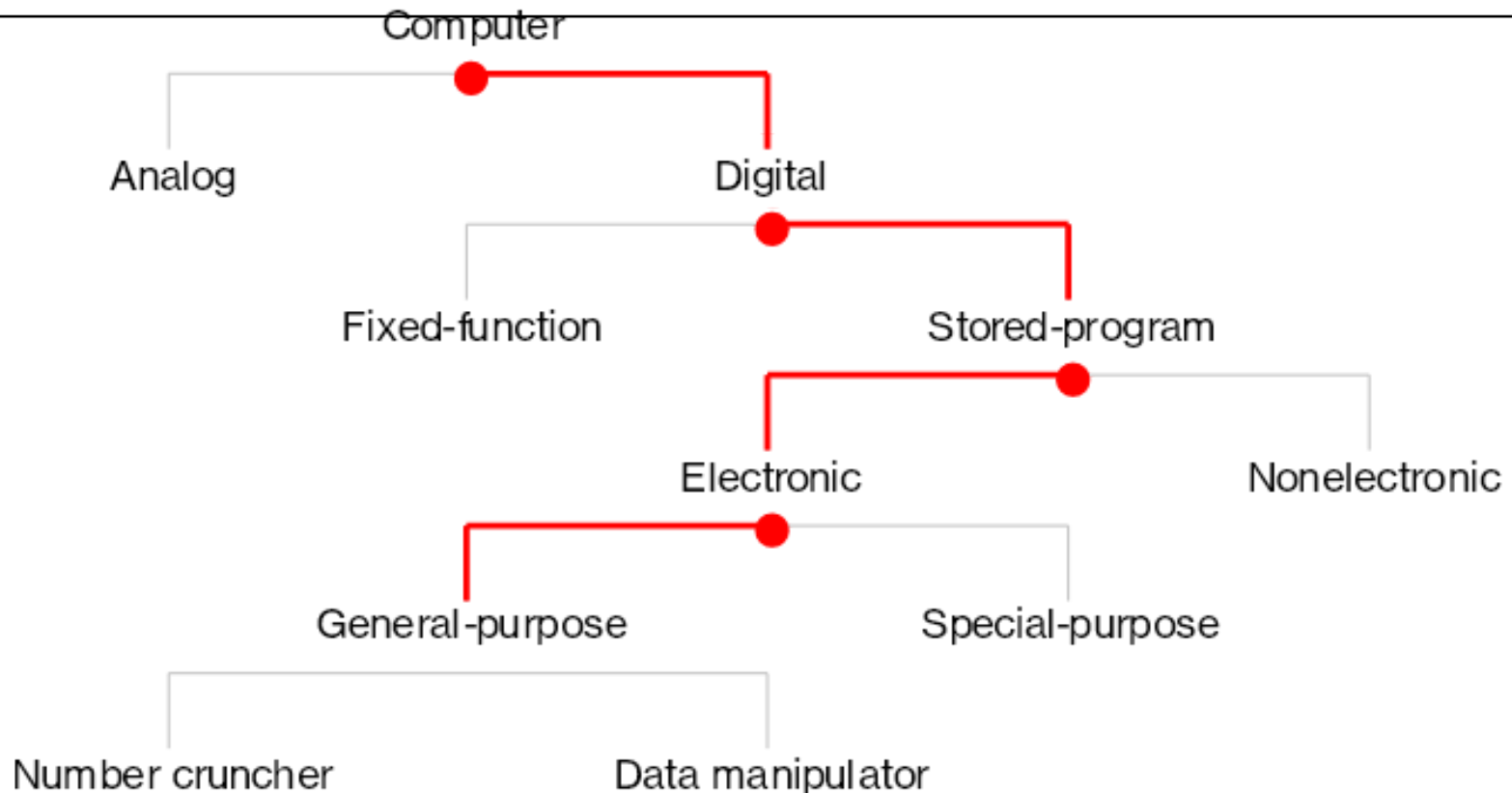
- How to integrate the components to build a computer system to achieve a desired level of performance.
- **Analogy:** architect's task during the planning of a building (overall layout, floorplan, etc.).

What Is (Computer) Architecture?



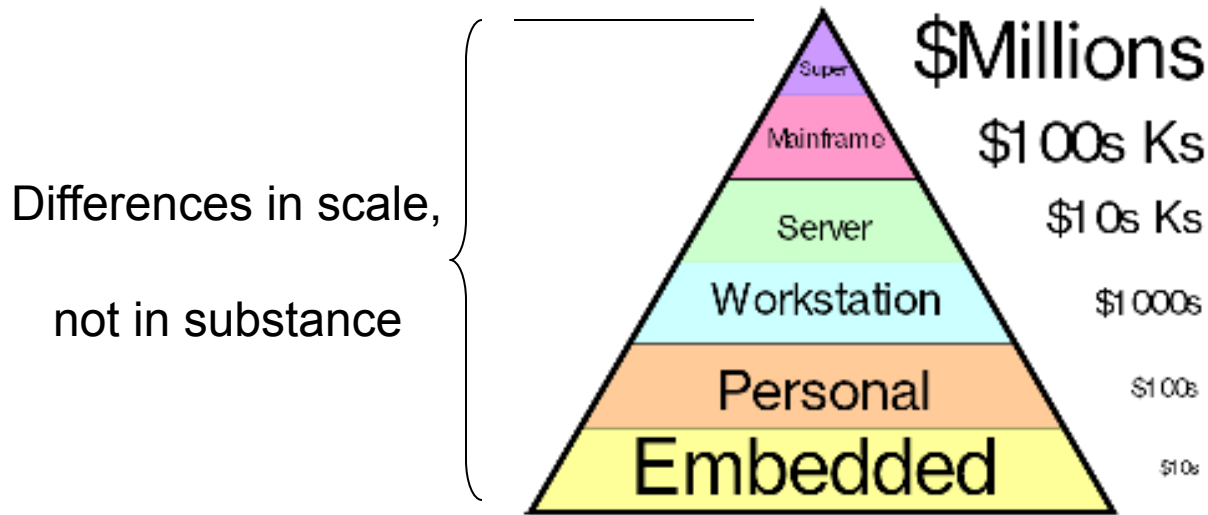
Like a building architect, whose place at the engineering/arts and goals/means interfaces is seen in this diagram, a computer architect reconciles many conflicting or competing demands.

3.2 Computer Systems and Their Parts



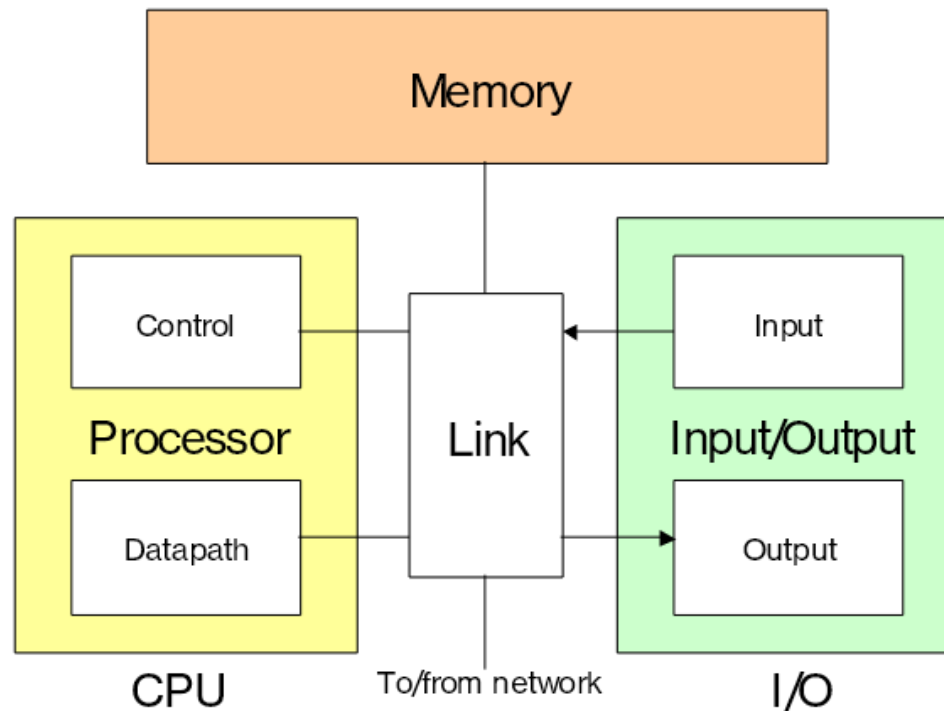
The space of computer systems, with what we normally mean by the word “computer” is highlighted.

Price/Performance Pyramid



Classifying computers by computational power and price range.

Digital Computer Subsystems



The (three, four, five, or) six main units of a digital computer. Usually, the link unit (a simple bus or a more elaborate network) is not explicitly included in such diagrams.

3.3 Generations of Progress

The 6 generations of digital computers, and their ancestors.

Generation (begun)	Processor technology	Memory innovations	I/O devices introduced	Dominant look & fell
0 (1600s)	(Electro-) mechanical	Wheel, card	Lever, dial, punched card	Factory equipment
1 (1950s)	Vacuum tube	Magnetic drum	Paper tape, magnetic tape	Hall-size cabinet
2 (1960s)	Transistor	Magnetic core	Drum, printer, text terminal	Room-size mainframe
3 (1970s)	SSI/MSI	RAM/ROM chip	Disk, keyboard, video monitor	Desk-size mini
4 (1980s)	LSI/VLSI	SRAM/DRAM	Network, CD, mouse, sound	Desktop/ laptop micro
5 (1990s)	ULSI/GSI/ WSI, SOC	SDRAM, flash	Sensor/actuator, point/click	Invisible, embedded

Source: Computer Architecture, Background and Motivation

An interesting view point on Architecture in the AI Age

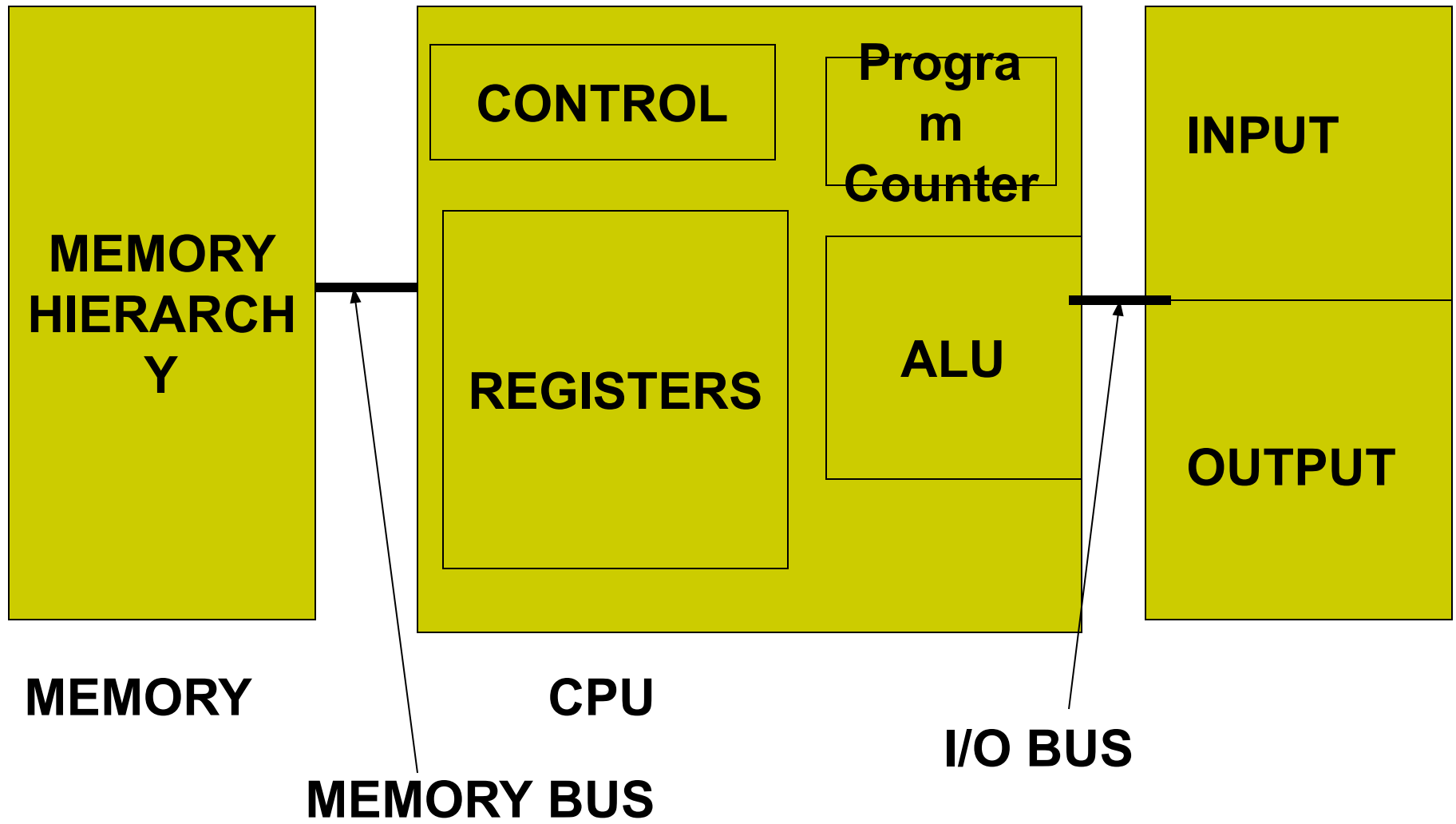
- <https://www.youtube.com/watch?v=Bf0hMhZ4qpQ>
 - Start at 14:35

Von Neumann Model



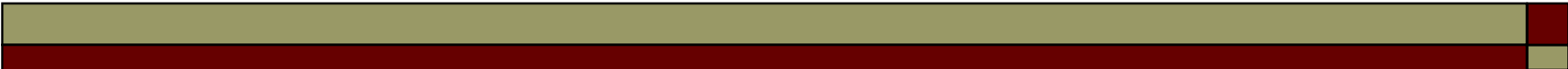
- 1903-1957
- Contributed to give a very basic model, often referred to as **Von Neumann model**

The von Neumann Machine



The Stored Program Concept

- Programs are sequence of instructions stored in the memory.
- The CPU consists of:
 - **Program Counter** (PC) indicates the address of the next instruction to be executed.
 - **ALU (Arithmetic Logic Unit)**
 - performs arithmetic and logical operations
 - **Registers:** High Speed storage of operands.
 - **Control Unit:** That interprets instructions and causes them to be executed.



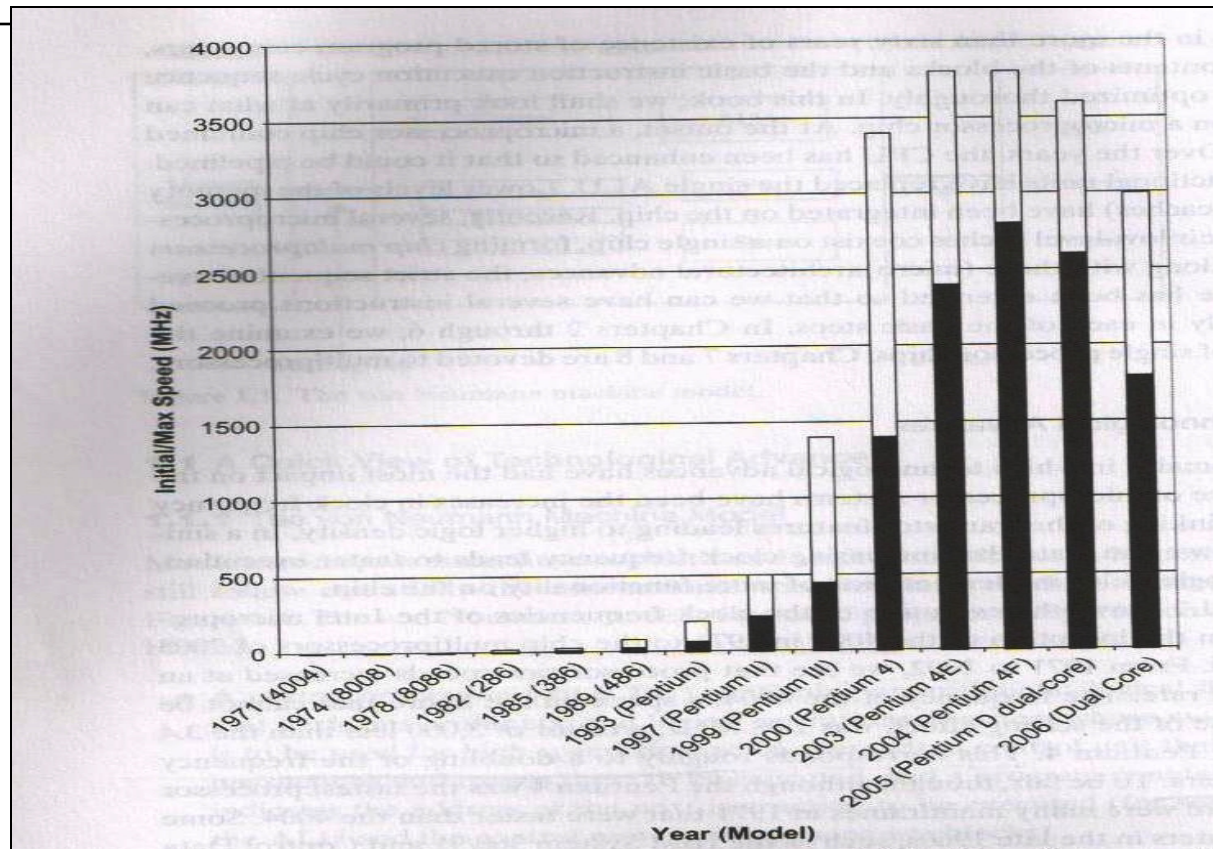
The Stored Program Concept

- The memory stores the instructions, data and intermediate results. Memory has several hierarchy:
 - registers being the highest level and the fastest form.
- Input/ Output: Transmits and receives results and messages (information) from and to the outside world respectively.

Instruction Execution Cycle

- Next instruction is fetched from memory.
- Control Unit decodes the instruction.
- Instruction is executed:
 - ALU based
 - **load** from a memory location to a register
 - **store** from a register to the memory
 - testing condition of a potential branch
- PC is updated (incremented except for a branch)
- Repeat

Brief History

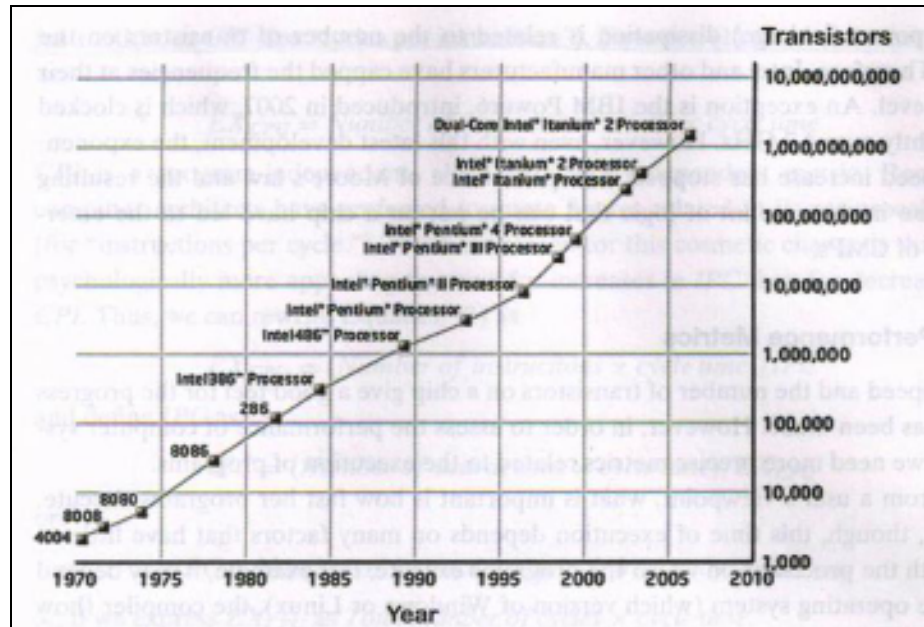


Black bars:
frequency at
introduction

White bars:
peak
frequency

- Evolution of Intel Microprocessors (from Jean Baer, Microprocessor Architecture, Cambridge University Press)

Moore's law for Intel Microprocessors



- Evolution of Intel Microprocessors (from Jean Baer, Microprocessor Architecture, Cambridge University Press)

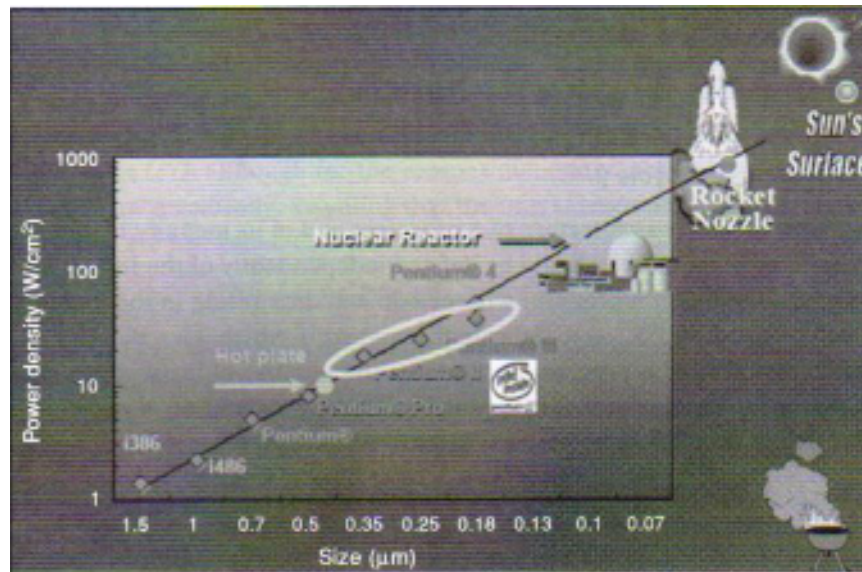
Some figures

- 1971: Intel 4004, 1.08 MHz, 2,300 transistors
- 2003: Intel Pentium 4, 3.4 GHz, 1.7 billion transistors
 - Frequency increases roughly double per 2.5 years
 - Number of transistors roughly double every two years (Moore's Law).
- How will the trend continue in the future?

Some figures

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 - Frequency increases roughly double per 2.5 years
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- How will the trend continue in the future?
 - IBM builds first sub 1 nm chip with 100 B transistors.
 - <https://backendnews.net/ibm-builds-first-sub-1-nm-chip-with-100b-transistors/?shem=dsdf,sharefoc,agadiscoverSDL,,sh/x/discover/m1/4>

Power Dissipation in Intel Processors



- Evolution of Intel Microprocessors (from Jean Baer, Microprocessor Architecture, Cambridge University Press)

Performance Metrics

- Raw Speed and number of transistors give a good indication of the developments made by the processors.
- However we need more precise metrics.
 - to evaluate how fast a program executes
- But it depends on several factors:
 - Operating System
 - Compiler
 - Network
 - Nature of Program

Program Independent Metrics and Benchmarks

- Program Independent metrics: Metrics which are not affected by the types of programs.
- Benchmarks: A suite of programs, that indicate the load of the processor.

Instructions Per Cycle (IPC)

- Metrics to assess the micro-architecture and the memory hierarchy, should be independent of the IO subsystem.
- Define, EX_{CPU} as the time to execute a program (a collection of instructions), when the code and data both reside in the memory.
 - $EX_{CPU} = \text{Number of Instructions} \times \text{Time to execute one instruction}$

CPI vs IPC

- Now, Time to execute one instruction = Cycles Per Instruction (CPI) x Cycle time
 - CPI is a program-independent, clock frequency independent metric.
- IPC (Instructions per cycle) = $1/\text{CPI}$
 - More intuitive as one tries to increase IPC (rather than decreasing CPI).

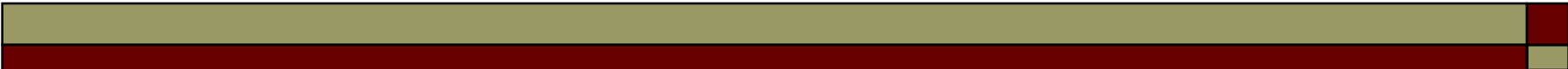
$$\text{IPC} = (\text{Number of Instructions} \times \text{cycle time}) / \text{EX}_{\text{CPU}}$$

Performance

- Can be defined as the reciprocal of EX_{CPU} .
- Three important factors:
 - Compiler driven: For a given set of Instructions (called as ISA-Instruction Set Architecture), and a given program, it decides the number of instructions.
 - Micro-architecture design and implementation: Smaller the CPI or more the IPC, better the performance.
 - Technology: Decides the cycle time.

Reduction of Instructions does not necessarily make the programs faster

- Consider the example, where a multiplication program can be realized by shift and add.
- The number of instructions increase.
- But the overall time required reduces.
 - This is because, not all instructions take the same number of cycles.



Conclusions

- Von Neumann Model: The Load Store Architecture
- The Instruction Cycle.
- The trend in processor designs, Moore's law.
- Performance Metrics.